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Section 8.2: The Analysis Stage

'**Analysis**' is the first stage tackled during systems analysis and design.

We will look at what is involved when analysing the current system.

Key Concepts of this section:

- # Understand what is involved in the analysis stage.
- # Be able to describe the steps that take place during the analysis stage.

Stage 1 - Analysis

What is analysis?

- # This is where the **current system is looked at in detail** in order to figure out what changes need to be made to make the new system better than the old one.

If there is no current system, analysis will look at the requirements of the proposed new system.

For example:

If a company does not have a current payroll system, the systems analysts will look closely at a list of features that management want the new system to have.

What does analysis involve?

- # Typical analysis would involve the following:



Key Words:

Analysis, Fact Finding, Cost-Benefit, Questionnaire, Interview, Observe

Examples:



A systems analyst looks at a current system (or a proposed system) in detail and decides

- 1 Collecting data about the current system / proposed system**
(what needs changing, what is working well etc)
- 2 Find out problems with the current system**
- 3 Establish the problem that the customer needs solving**
(What does the new system need to be able to do? - e.g. calculate employee pay)
- 4 Identify inputs, processing and outputs of the current / proposed system**
(What needs to go into the system?, what calculations?, what needs to come out?)
- 5 Identify the requirements of the new system**
(What tasks should the system be able to handle?)
- 6 Producing a cost-benefit analysis**
(Is it worth it financially to build the system?).

How is this data obtained?

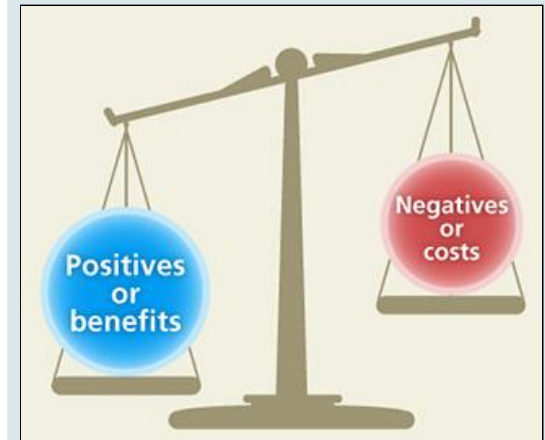
- # Finding out information about the current / proposed system is known as '**fact finding**'.
- # There are four methods which are used to obtain this information:
 - **Questionnaires**
 - **Interviews**
 - **Observing people using the current system**
 - **Looking at the current system's documentation**

These methods will be discussed below:

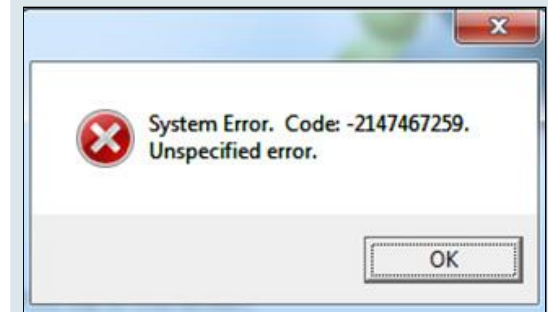
1 Questionnaires

- # Prepared questions are given to users of the system and they are **left with the user to complete**.
- # Questionnaires usually focus on more **simple questions** and are completed by **ticking or circling options** or **shading boxes**.
- # The questionnaire will contain questions that are designed to extract useful information about the current system / proposed system.

how to make it work better.



Analysis can determine if it is financially worth it to implement a new system.



Analysis can help identify errors and faults with the current system.

Examples:

For example:

- * Is the current system easy to use? **YES/NO**
- * Does the current system freeze or crash? **YES/NO**
- * How fast does the current system handle tasks? **SLOW/FAST/VERY FAST**

Advantages / Disadvantages of questionnaires:

Advantages	Disadvantages
Questions can be answered quickly .	People often do not complete or return the questionnaire.
Answers are more honest as the questions can be answered anonymously .	Unclear questions cannot be explained as people are left to answer the questions alone.
Fairly cheap method of gathering data.	You may get incorrect data if people have misunderstood a question .
Answers to the questionnaires can be analysed automatically using an OMR (optical mark reader).	It is hard to ask very technical or specific questions on a questionnaire.



Questionnaires are quick ways of gathering basic information about a current system.



Questionnaire findings can be analysed quickly using optical mark readers.

2 Interviews

- # Interviews take place **face-to-face** and usually **involve more detailed questions** than questionnaires.
- # The interviewer talks to **people at various levels of the business** (managers, directors, employees etc).
- # Different questions might be asked to different employees.

For example:

- * **A manager/director** might be asked questions focused on the **exact requirements** of a new system
- * **An employee** might be asked questions about the **current system, how they use it** and **any problems** that they might have.

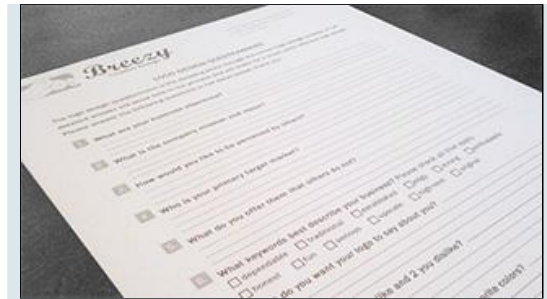
Advantages / Disadvantages of Interviews:

Examples:



Interviews take place face-to-face .

Advantages	Disadvantages
Questions can be explained if they are misunderstood.	Interviews take far longer to complete than questionnaires.
More complex questions can be asked which will give more detailed findings .	Expensive to carry out as the person being interviewed needs to be taken away from their work.
Questions can be changed to suit who is being interviewed (different questions for a director compared to a regular employee).	Answers may not be honest as the person being interviewed cannot remain anonymous (they may give answers they think are expected rather than the truth).
You will get a full set of data as the person being interviewed has no choice but to answer all questions.	



Interviews allow for more complex questions to be asked. This often gives more detailed information.

3 Observations

- # This is where a **systems analyst sits and watches** somebody using the current system.
- # By watching the current system being used, the analyst can **log or make notes about different facts**:

For example:

- * What are the **inputs, processes** and **outputs**?
- * What does the system **do well**?
- * Are there **any errors or faults** with the current system?

Advantages / Disadvantages of Observations:

Advantages	Disadvantages
Analyst can see exactly what the current system does well and not so well .	Person being watched might feel uncomfortable and work in a different way to usual.
Not expensive to carry out as the employee is not taken away from their work.	

Examples:



Observing employees using the system gives the analyst an exact picture of what the current can do.

This method of gathering information about the system is very cheap as the employee is not taken away from their work.

Looking at the current system's documentation

This involves looking at **paperwork** for the current system. The paperwork will contain **information needed to implement the new system**.

Paperwork can include:

- * Lists of **stock items** that will be **stored in the new system**
- * **Employee pay scales** (needed for processing in a payroll system)
- * **Technical documentation** for the current system which could help develop improvements for the new system.

For example:

Technical documentation explains exactly **how the current system has been built**. This helps developers implement good features of the old system into the new one.

Examples:



Existing system documentation can help analysts understand how the current system has been built.



Lots of paperwork is very time consuming and expensive to look through.

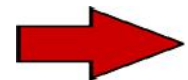
Advantages / Disadvantages of looking at existing documentation:

Advantages	Disadvantages
Could save time as there may be copies of previous analysis.	Very time consuming to look through all of the existing documents.
Can see existing inputs, processing and outputs .	Very expensive as the analyst will need to be paid for time spent looking at documentation.
Allows the analyst to predict the size of the system needed by looking at the amount of data that it will be required to handle.	Time could be wasted if existing documentation is not relevant to the new system.



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